

Cursor AI Instructions: Upgrading the Mon Valley Pollution Tracking System

Introduction

This document provides specific, actionable instructions for using Cursor AI to upgrade the Mon Valley Pollution Tracking System to match the design quality and user experience of ClimateIQ. These instructions are based on the technical roadmap and should be executed sequentially.

Phase 1: Foundation Enhancement

Goal: Modernize the project setup for better performance and developer experience.

1.1: Migrate from Create React App to Vite

Cursor AI Prompt:

Analyze the existing Create React App project structure. Create a new Vite project with the React + TypeScript template. Then, migrate all existing components, assets, and dependencies from the CRA project to the new Vite project. Ensure all import paths are updated and the application runs correctly with the Vite development server.

1.2: Implement Tailwind CSS with a Custom Theme

Cursor AI Prompt:

Integrate Tailwind CSS into the Vite project. Create a `tailwind.config.js` file and configure a custom theme based on the design of climateiq.org. Define a professional color palette with dark and light mode support, a modern typography scale with appropriate font sizes and line heights, and a consistent spacing system. Apply Tailwind's base styles to the entire application.

1.3: Create a Reusable Component Library

Cursor AI Prompt:

Create a new directory `src/components/ui`. For each of the following UI elements, create a new, reusable React component using Tailwind CSS for styling. Ensure the components are accessible and follow best practices. The components to create are: `Button`, `Card`, `Input`, `Select`, `Checkbox`, `Modal`, and `Tooltip`.

Phase 2: Animation & Interaction Layer

Goal: Introduce smooth animations and micro-interactions using Framer Motion.

2.1: Integrate Framer Motion for Page Transitions

Cursor AI Prompt:

Install the `framer-motion` library. In the main App component, use the `AnimatePresence` component to wrap the React Router switch. Then, for each page component, add a `motion.div` with `initial`, `animate`, and `exit` props to create a smooth fade-in/fade-out transition between page navigations.

2.2: Add Scroll-Triggered Animations

Cursor AI Prompt:

For each major section on the home page, create a scroll-triggered animation using Framer Motion's `useInView` hook. As the user scrolls down, each section should fade in and slide up into view. Apply this effect to the "About," "Mission," and "Features" sections.

Phase 3: Visual Design Overhaul

Goal: Redesign the application's visual appearance to be modern and professional.

3.1: Redesign the Navigation and Header

Cursor AI Prompt:

Redesign the main navigation bar to be a sticky header that remains at the top of the page on scroll. Replace the emoji icons with professional icons from the `lucide-react` library. The header should have a blurred background effect and a subtle shadow. The navigation links should have a smooth hover effect.

3.2: Implement a Modern Hero Section

Cursor AI Prompt:

On the home page, create a new hero section similar to climateiq.org. It should feature a large, bold headline, a short descriptive paragraph, and a primary call-to-action button. Use a high-quality background image with a subtle parallax effect on scroll.

Phase 4: Layout & Composition Refinement

Goal: Improve the page structure and layout for a more engaging user experience.

4.1: Create Split-Screen Layouts

Cursor AI Prompt:

On the home page, refactor the "Features" section into a series of split-screen layouts. For each feature, display the text description on one side and a relevant visual or icon on the other. Alternate the layout (text-left, visual-right, then text-right, visual-left) for each feature.

Phase 5: Data Visualization Enhancement

Goal: Improve the visual quality and interactivity of charts and maps.

5.1: Redesign the Leaflet Map

Cursor AI Prompt:

Customize the styling of the Leaflet map on the "Sensor Map" page. Use a modern, dark-themed map tile layer from Mapbox or a similar provider. Design new custom map markers with subtle animations on hover. The map controls should also be restyled to match the application's new design system.

5.2: Animate the Data Charts

Cursor AI Prompt:

On the "Dashboard" page, use the `recharts` library to rebuild the PM2.5 trend chart. The chart should animate in when it becomes visible. The tooltips should be redesigned to be more informative and visually appealing. The chart's colors should align with the new application-wide color palette.

Phase 6: Performance & Polish

Goal: Optimize the application for speed, accessibility, and a polished final product.

6.1: Implement Lazy Loading and Code Splitting

Cursor AI Prompt:

Use `React.lazy` and `Suspense` to implement route-based code splitting. Each page component should be loaded lazily. Also, lazy-load any large components or libraries that are not needed for the initial render. Add a simple loading spinner as the fallback UI for the `Suspense` component.

6.2: Ensure Accessibility (WCAG 2.1 AA)

Cursor AI Prompt:

Analyze the entire application for accessibility issues. Ensure all images have `alt` tags, all form inputs have labels, and all interactive elements are keyboard-navigable. Use the `eslint-plugin-jsx-a11y` ESLint plugin to identify and fix accessibility violations. Ensure the color contrast ratios meet WCAG 2.1 AA standards.